Ch 3 · Undifferentiated shock

Four types

- Hypovolaemic and cardiogenic: low CO, high SVR (cardiogenic has high filling pressures)
- Obstructive (tamponade, PE, tension): low CO, raised CVP; distributive: high CO, low SVR

Bedside assessment (RUSH)

- Pump: LV/RV function, effusion; Tank: IVC, FAST, B-lines, pleura; Pipes: aorta, DVT
- Lactate, ScvO2, CO2 gap, capillary refill, mottling, urine output

Resuscitation

- Fluid responsiveness: PLR with real-time CO (≥ 10%), PPV if ventilated in sinus rhythm
- 250–500 mL boluses with reassessment; early noradrenaline to MAP ≥ 65
- Treat cause in parallel: antibiotics/source, drainage, reperfusion, thrombolysis

Key point: Ultrasound first, fluids second – shock type decides treatment.

Ch 20 · Severe community-acquired pneumonia

Severity

- CURB-65 ≥ 3 severe; IDSA/ATS major: vasopressors or ventilation; ≥ 3 minor criteria

Antimicrobials

- Beta-lactam + macrolide (or respiratory quinolone); within 1 h if shock
- Post-influenza/necrotising: cover S. aureus incl. MRSA (linezolid); oseltamivir for influenza
- Tests: cultures, urinary pneumococcal/Legionella antigens, viral PCR; 5–7 days if improving

Corticosteroids

- CAPE COD (2023): hydrocortisone 200 mg/day ↓ 28-day mortality in severe CAP
- Not for influenza pneumonia; watch glucose, infection

Key point: Severe CAP answer: severity → antibiotics → steroids → oxygen/ventilation strategy.

Ch 24 · Pneumonia in the immunocompromised

Differential by host

- HIV CD4 < 200: Pneumocystis; neutropenia: Gram-negatives, Aspergillus; transplant: CMV, fungi, Nocardia
- Non-infective: drug toxicity, lymphoma, Kaposi, DAH, IRIS

Pneumocystis (PJP)

- Induced sputum/BAL PCR; beta-D-glucan supportive; LDH raised
- Co-trimoxazole high dose 21 days; prednisolone if PaO2 < 9.3 kPa
- Alternatives: primaquine + clindamycin, IV pentamidine

ART timing

- Start within 2 weeks of PJP treatment; watch for IRIS
- Delay ART in cryptococcal/TB meningitis

Key point: Early BAL changes management in most immunocompromised patients.

Ch 29 · Ventilator-associated pneumonia

Diagnosis

- Pneumonia > 48 h after intubation: new infiltrate + fever/WBC/purulence/worse oxygenation
- Lower airway sample before antibiotics (ETA semi-quantitative or BAL ≥ 10^4 CFU/mL)

Empirical treatment

- Anti-pseudomonal beta-lactam; add MRSA cover if local MRSA > 10–20%
- Second Gram-negative agent if shock or high resistance; de-escalate; 7 days

Prevention bundle

- Head up 30–45°, sedation holds + SBT, subglottic drainage, cuff 20–30 cmH2O
- Avoid circuit changes, early extubation; SDD in low-resistance units (SuDDICU)

Key point: Shorter courses (7 days) are as effective and reduce resistance.

Ch 54 · Adrenal insufficiency and crisis

Types

- Primary: autoimmune, TB, haemorrhage, metastases – low aldosterone (high K+)
- Secondary: pituitary disease, steroid withdrawal (commonest in ICU)

Adrenal crisis

- Hypotension refractory to fluids, vomiting, confusion, low Na, high K, hypoglycaemia
- Hydrocortisone 100 mg IV now, then 200 mg/24 h; saline 1 L in first hour; glucose
- Take cortisol/ACTH first only if it does not delay treatment

Prevention

- Sick-day rules, emergency injection kit, steroid card; peri-operative cover

Key point: Never wait for a cortisol result before giving hydrocortisone in suspected crisis.

Ch 55 · CIRCI and steroids in septic shock

Concept

- Inadequate corticosteroid activity for illness severity: HPA dysfunction, reduced cortisol clearance, tissue resistance
- Random cortisol/ACTH test unreliable in septic shock

Evidence

- ADRENAL: no mortality benefit, faster shock reversal
- APROCCHSS: hydrocortisone + fludrocortisone ↓ 90-day mortality; CORTICUS: no benefit
- SSC 2021: hydrocortisone 200 mg/day if noradrenaline ≥ 0.25 µg/kg/min for ≥ 4 h

Adverse effects

- Hyperglycaemia, hypernatraemia, weakness, infection, GI bleeding, delirium

Key point: Steroids in septic shock reverse shock faster; mortality benefit is inconsistent.

Ch 59 · Urosepsis

Approach

- Cultures then antibiotics within 1 h; ESBL risk → carbapenem ± amikacin
- Ultrasound/CT for obstruction, abscess, emphysematous pyelonephritis

Source control

- Obstructed infected kidney = emergency decompression
- Nephrostomy (no anaesthesia) or retrograde stent – equivalent; stone treatment later

Resistance risk

- Recent antibiotics/hospitalisation, long-term catheter, instrumentation
- Nursing-home residence, prior ESBL/CRE, travel to high-prevalence areas

Key point: Pus under pressure needs drainage – antibiotics alone fail.

Ch 60 · Sepsis and septic shock

Sepsis-3

- Sepsis: infection + organ dysfunction (SOFA ≥ 2); shock: vasopressor for MAP 65 + lactate > 2 after fluids
- qSOFA is prognostic, not a screening tool

Hour-1 bundle

- Lactate, cultures, broad-spectrum antibiotics, 30 mL/kg crystalloid if hypoperfused, noradrenaline
- Vasopressin when noradrenaline 0.25–0.5 µg/kg/min; source control within 6–12 h

Evidence

- EGDT trials (ProCESS, ARISE, ProMISe): no benefit; balanced crystalloids (SMART), avoid starch
- CLASSIC/CLOVERS: restrictive = liberal; ANDROMEDA-SHOCK: capillary refill targeting

Key point: Time to antibiotics and source control drives outcome.

Ch 61 · Invasive aspergillosis

Risk factors

- Prolonged neutropenia, haematological malignancy, HSCT, GVHD, high-dose steroids
- ICU: COPD on steroids, cirrhosis, severe influenza/COVID-19

Diagnosis

- CT: halo sign (early), air crescent (late), cavitation
- Galactomannan (serum/BAL), Aspergillus PCR, BAL culture (septate hyphae at 45°)

Treatment

- Voriconazole or isavuconazole first-line; liposomal amphotericin if mucormycosis possible
- TDM for voriconazole; reduce immunosuppression; ≥ 6–12 weeks

Key point: Influenza-associated aspergillosis: send BAL galactomannan in ventilated patients.

Ch 62 · Strongyloides hyperinfection

Mechanism

- Autoinfection persists for decades; steroids/HTLV-1 → massive larval migration
- Larvae carry gut bacteria → Gram-negative bacteraemia/meningitis; eosinophilia often absent

Diagnosis

- Larvae in sputum/BAL/stool (agar plate culture); serology (false negatives)

Management

- Ivermectin daily until negative for 2 weeks (subcutaneous if ileus)
- Reduce steroids; antibiotics for enteric organisms
- Screen/treat people from endemic areas before immunosuppression

Key point: Unexplained Gram-negative sepsis on steroids in someone from the tropics – think Strongyloides.

Ch 63 · Viral haemorrhagic fevers

Viruses

- Ebola/Marburg (body fluids), Lassa (rodents), CCHF (ticks), dengue, yellow fever
- Incubation ≤ 21 days; exclude malaria first

Infection control

- High-level isolation, notify public health, full PPE with trained donning/doffing buddy
- Limit procedures/samples, dedicated equipment, waste protocols

Treatment

- Aggressive fluids and electrolytes; organ support improves survival
- Ebola Zaire: monoclonal antibodies (PALM); Lassa: ribavirin

Key point: Staff safety first – no exposure-prone procedure without full PPE.

Ch 64 · Severe malaria

Severe falciparum criteria

- Impaired consciousness, seizures, prostration, parasitaemia > 10% (> 2–5% non-immune)
- Hypoglycaemia, lactate > 5, Hb < 7, AKI, jaundice, ARDS, shock, DIC

Treatment

- IV artesunate 2.4 mg/kg at 0, 12, 24 h then daily (AQUAMAT, SEAQUAMAT)
- Then full oral ACT; watch delayed haemolysis
- Glucose, cautious fluids, RRT, seizure control, cover bacteraemia; no exchange transfusion

Key point: Artesunate beats quinine – give it without delay.

Ch 65 · Dengue

Phases

- Febrile → critical (defervescence, 24–48 h plasma leak) → recovery (fluid reabsorption)

Warning signs / severe dengue

- Abdominal pain, vomiting, mucosal bleeding, lethargy, rising haematocrit with falling platelets
- Severe: shock (pulse pressure ≤ 20), respiratory distress from leak, severe bleeding, organ failure

Management

- Crystalloid 5–10 mL/kg/h titrated to haematocrit, vitals, urine; colloid if refractory
- Reduce fluid as leak ends; falling haematocrit + instability = bleeding → transfuse
- No NSAIDs/IM injections; no prophylactic platelets

Key point: Danger is the afebrile critical phase – fever settling is not recovery.

Ch 66 · Leptospirosis

Features

- Rodent-urine water exposure; fever, calf myalgia, conjunctival suffusion
- Weil's disease: jaundice + AKI + bleeding; pulmonary haemorrhage = highest mortality

Diagnosis

- PCR blood (week 1) and urine (later); IgM ELISA, MAT
- Hypokalaemic non-oliguric AKI, high CK, low platelets, conjugated bilirubin

Treatment

- Penicillin, ceftriaxone or doxycycline; Jarisch–Herxheimer reaction possible
- Early daily dialysis, lung-protective ventilation, ECMO for haemorrhage

Key point: Jaundice with high CK and hypokalaemic AKI after flooding = leptospirosis.

Ch 72 · Clostridioides difficile infection

Severity

- Non-severe: WBC < 15, creatinine < 133; severe: WBC ≥ 15 or creatinine > 133
- Fulminant: shock, ileus, megacolon

Treatment

- Non-severe/severe: fidaxomicin or oral vancomycin 10 days
- Fulminant: oral/NG vancomycin 500 mg qds + IV metronidazole ± rectal vancomycin
- Early surgery: subtotal colectomy or loop ileostomy + lavage

Control and recurrence

- Contact precautions, soap and water, sporicidal cleaning; stop PPIs/unneeded antibiotics
- Bezlotoxumab, faecal microbiota transplant for recurrences

Key point: Rising lactate or WBC > 25 in fulminant CDI = surgical review now.

Ch 87 · Complicated intra-abdominal infection

Classification

- Uncomplicated vs complicated; community vs healthcare-associated (resistant)
- Primary (SBP), secondary (perforation, leak), tertiary (persistent)

Management

- Source control within 6–12 h (relaparotomy, stoma, percutaneous drainage)
- Healthcare-associated: pip-taz or carbapenem ± antifungal ± enterococcal cover
- Peritoneal cultures, de-escalate

Duration

- STOP-IT: 4 days after adequate source control

Key point: Source control is the treatment; antibiotics are adjuncts.

Ch 89 · Abdominal and extremity compartment syndrome

Definitions

- IAH ≥ 12 mmHg (grades I–IV); ACS = IAP > 20 with new organ failure
- APP = MAP − IAP, aim > 60

Effects

- Oliguria, high airway pressures, low venous return/CO, raised ICP, gut ischaemia

Management

- Sedation, paralysis, NG/rectal decompression, drain ascites, negative fluid balance
- Decompressive laparotomy if refractory
- Limb compartment syndrome: measure pressures, fasciotomy

Key point: Measure bladder pressure in every high-risk patient – clinical exam is unreliable.

Ch 91 · Major burns

Airway and inhalation

- Early intubation if stridor, hoarseness, facial burns, soot
- 100% O2 for CO; hydroxocobalamin for suspected cyanide

Fluids

- Parkland: 4 mL × kg × %TBSA; half in first 8 h from burn
- Titrate to urine 0.5 mL/kg/h; avoid fluid creep; albumin rescue

Ongoing care

- Escharotomy for circumferential burns; early excision and grafting
- Hypermetabolism: early enteral feeding, high protein, propranolol, oxandrolone; infection control

Key point: Formula is a starting point – urine output decides the rate.

Ch 93 · Necrotising soft tissue infection

Recognition

- Pain out of proportion, spreading erythema, bullae, crepitus, anaesthesia, shock
- Types: I polymicrobial, II group A strep, III clostridial/Vibrio; LRINEC supportive only

Management

- Immediate surgical debridement, re-look within 24 h
- Pip-taz/carbapenem + vancomycin/linezolid + clindamycin (toxin suppression)
- Resuscitation; IVIG for streptococcal toxic shock

Imaging

- Do not delay surgery for CT when suspicion is high

Key point: Every hour of delayed debridement raises mortality.